# Malengo Savings Plan

Product guide and configuration reference. Prepared 15 October 2026.

## 1. What the plan is

A savings contract. The client pays in whenever they like and takes money out whenever they like, and the balance earns 4% a year, credited at each month end. There is no fixed sum assured: what the client is owed is whatever the account holds. Taking the whole balance out is how the contract ends.

## 2. The configurable part

Set on the product version when it is published. Changing any of these means publishing a new version; a policy already sold keeps the version it was sold on.

| Setting | Malengo | What it does |
|---|---|---|
| guaranteedRatePercent | 4 | Annual rate floor, credited monthly |
| minimumBalance | 0 | A withdrawal may empty the account |
| contributionAllocation% | 0 | Held back from each deposit |
| transferAllocation% | 0 | Held back from a transfer in |
| monthlyPolicyFee | 1,000 | Taken at each month end |
| minEntryAge / maxEntryAge | 18 / 60 | Enforced when a case is opened |
| payout amountBasis | ACCOUNT_VALUE | Maturity pays the account, not a face amount |
| portfolioCode | SAV | An account must be a savings contract |

The minimum balance is the one setting that separates Malengo from the Akiba plan beside it: Akiba keeps a 50,000 floor, so its account can never be emptied. Malengo's is zero.

## 3. Setting it up

1. **Author the product: category ENDOWMENT, portfolio SAV, currency TZS.**
2. **Publish a version carrying the settings in section 2, with a rating table covering ages 18 to** 60 and a MATURITY payout row on ACCOUNT_VALUE.
3. **Open an underwriting case for the client and decide it. Entry age is checked here.**
4. **Issue the policy. The sum assured carries the opening deposit -- see section 5.**
5. **Collect the first premium. That is the opening deposit, and it opens the account.**

Steps 1 and 2 are scripted in backend/scripts/malengo-demo.sh.

## 4. Running it

A deposit is a premium collection or an ad-hoc top-up; both credit the account in full, since nothing is held back. A withdrawal is asked for by one person and approved by a second, then paid through the payment rail -- and a second withdrawal is refused while one is still in flight. Every movement lands on an append-only ledger: a correction is posted as a reversal, never an edit. At each month end the engine credits interest, then takes the policy fee.

## 5. Two things to know

- No fixed sum assured, but not a zero one. The wire layer refuses a sum assured of zero, so the contract carries the opening deposit as its face amount. It is a placeholder, not a promise: the maturity payout is ACCOUNT_VALUE. This is the convention the fixed-term deposit already uses.
- Emptying the account does NOT yet end the policy. A withdrawal that takes the balance to zero leaves the policy ACTIVE and the account OPEN, because only the month-end charge run closes an account by exhaustion. Until that is built, a Malengo contract is ended by SURRENDER, which pays the balance and closes the policy properly.

## 6. What a year looks like

Simulated against a real database through the real engine (MalengoSimulationTest), with every date backdated. A 35-year-old client, 1,000,000 paid in across the year, 150,000 out.

| Step | Balance (TZS) |
|---|---|
| Opening deposit 500,000, then 2 month-ends | **501,284.88** |

| | |
|---|---|
| Top-up 200,000, then 3 more month-ends | **704,418.86** |
| Withdrew 150,000, then 3 more month-ends | **558,407.20** |
| Deposited 300,000, then the rest of the year | **866,715.83** |
| Withdrew the whole balance | **0.00** |

850,000 of net cash in became 866,715.83: 4% compounded monthly, less twelve 1,000 fees. After the final withdrawal the policy was still ACTIVE on a zero balance, which is the gap section 5 describes.